FinTwin: A Point-in-Time System-Dynamics Digital Twin for Corporate Liquidity Stress Testing from SEC Filings

Spilios Dimakopoulos

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Abstract

We present FinTwin, an open-source digital twin of non-financial public companies that converts SEC XBRL filings into a calibrated stock-and-flow model of the firm's financial machinery and stress-tests it with Monte Carlo simulation. Unlike price-forecasting models, the twin simulates how cash, working capital, debt maturities, refinancing and credit quality jointly respond to shocks, through explicit feedback loops between interest coverage, synthetic ratings and market access. The data layer resolves three practical obstacles of XBRL — tag drift, year-to-date reporting and restatements — and produces point-in-time vintages that make backtests free of look-ahead bias. On 30,414 out-of-sample, four-quarter forecasts for 375 companies over 13 yearly cutoffs, the twin's revenue forecasts improve the continuous ranked probability score on a seasonal random walk by 28%; raw interval coverage is below nominal (70% for nominal 90%), and online conformal rescaling, learned only from earlier cutoffs, restores it to 91%. Calibrated only on filings available on 20 February 2020, the twin's stress health score ranks 331 companies' realized 2020–21 cash burn with Spearman $\rho = 0.39$ (95% bootstrap CI 0.29–0.48). A TypeScript port runs 5,000 paths in the browser and is parity-tested against the Python reference to 10^{-6}.

1 Introduction

Corporate stress testing asks a conditional question: *if* revenue falls 30% or debt markets close, does the company run out of cash, and how fast? Answering it requires a model of the firm's internal dynamics — operating leverage, working capital, the maturity wall, the cost and availability of refinancing — rather than a statistical forecast of a single output. System dynamics [1, 2] provides exactly this language: stocks that accumulate, flows that fill and drain them, and feedback loops whose dominance determines whether a shock fades or spirals.

Three gaps motivate this work. First, public corporate stress tests are either regulator-run for banks or proprietary. Second, open financial models built on SEC data rarely address the data's point-in-time nature, so their backtests silently use restated figures. Third, simulation models are seldom evaluated as probabilistic forecasts with proper scoring rules [3]. FinTwin contributes (i) a reproducible XBRL pipeline with point-in-time vintages, (ii) a stock-and-flow twin with explicit reinforcing and balancing loops, calibrated per company, (iii) a large-sample probabilistic evaluation with online conformal calibration [4, 5], and (iv) a cross-sectional test of whether the twin, frozen before COVID-19, identified the companies that would bleed cash.

2 Data: point-in-time XBRL

The SEC `companyfacts` endpoint returns every tagged value a company has filed. We construct a quarterly panel of 20 fields (income, cash flow, balance sheet, debt components, customer deposits and the debt maturity schedule) with three corrections.

1. **Tag drift.** Each field has an ordered list of fallback tags merged period by period (e.g. `SalesRevenueNet` → `RevenueFromContractWithCustomer...` after ASC 606). Debt is assembled from current, non-current, commercial-paper and short-term components; operating income is rebuilt as pre-tax income plus interest where untagged.

2. **Year-to-date de-cumulation.** 10-Q cash-flow facts are cumulative and Q4 exists only within the 10-K. Facts sharing a start date are differenced (H1−Q1, 9M−H1, FY−9M); 52/53-week fiscal calendars are aligned within ±12 days.
3. **Adjusted operating income.** Non-cash goodwill, intangible and long-lived asset impairments are added back, as credit analysts do; overlapping tags (a cash-flow total versus its components) are reconciled by taking the largest measurement per quarter. Without this step a single write-down (e.g. Kraft Heinz's $9B in one quarter) drives trailing coverage negative and the synthetic rating to default.
4. **Point-in-time vintages.** `load_company(ticker, asof)` keeps only facts *filed* on or before `asof`, and the latest such filing wins, reproducing an analyst's information set and excluding later restatements.

Interest rates and recession dates come from FRED. Every number retains the accession number of its source filing.

3 The twin

3.1 Stocks, flows and loops

The model is quarterly and vectorized over Monte Carlo paths. Its stocks are cash, receivables, inventory, payables, customer deposits, net PP&E, term debt, revolver and equity; a residual "other net assets" stock makes the opening balance sheet balance, and the identity $A - L - E = 0$ is preserved on every path and quarter (verified by property-based tests). Demand follows an AR(1) in deseasonalized log growth,

$$g_t = \mu + \phi(g_{t-1} - \mu) + \sigma_g\,(1 + 0.5\,s_t)\,\varepsilon_t^d + \delta\,s_t, \qquad \mathrm{Rev}_t = B_{t-1}e^{g_t}\,S_{q(t)}\,(1+\xi_t), \tag{1}$$

where $s_t \in \{0,1\}$ is a Markov recession regime with entry hazard and exit probability estimated from NBER recessions since 1947, $\delta = -\beta\,\Delta_{\mathrm{GDP}}$ scales the historical nominal-GDP shortfall by the company's revenue cyclicality β, and ξ_t is a scenario shock. Cash operating costs are $F(1+\pi)^t c_t + (v + m_t)\mathrm{Rev}_t$, with a mean-reverting margin shock m_t correlated with demand. Receivables and inventory adjust toward days-based targets with first-order delays; payables and customer deposits move immediately, so a revenue collapse triggers deposit refunds.

Four feedback loops govern the response to stress:

- **R1 leverage–spread spiral.** Trailing interest coverage maps to a synthetic rating and spread [8]; refinanced and new debt carry that spread, raising interest and lowering coverage.
- **R2 market-access cliff.** The share of a funding need that can be placed falls from 100% (coverage $\geq 1.5\times$) to 60% and 30%, and to zero in a market freeze; unfunded maturities drain cash.
- **B1 cash conservation.** Under stress, payouts stop and capex falls to maintenance.
- **B2 delayed restructuring.** After two quarters below the target margin, fixed costs fall 4% per quarter to a floor of −15%.

A *liquidity breach* occurs when cash is negative after the revolver, accessible debt markets and (optionally) equity. A 0–100 health score combines liquidity runway (30%), synthetic rating (30%), net debt/EBITDA (20%) and Altman's Z″ [7] (20%). Because leverage that would alarm a software company is normal for a regulated utility, the coverage and leverage components are scored against sector norms: each GICS sector receives a tolerance factor equal to its median net debt/EBITDA relative to the universe median, clipped to $[1, 2]$. The factor is estimated from the data and affects the score only, not the dynamics.

3.2 Calibration

All parameters are estimated from the company's own point-in-time history: seasonal factors by median ratio-to-moving-average; trend growth shrunk 50% toward 1.2% per quarter; volatilities from the median absolute deviation of AR(1) residuals, with Student-t tails fitted by maximum likelihood; the fixed/variable cost split by penalized least squares shrunk toward a 35% fixed share when revenue variation cannot identify it; the debt-rollover share from the two-year debt trend; and the minimum operating cash balance from the company's own revealed behaviour (half its 10th-percentile cash balance over three years). An earlier cost-proportional cash floor forced high-turnover, low-cash retailers to "borrow up" to cash levels they never hold; the large-sample evaluation exposed this bias, and the revealed floor cut the twin's cash-forecast CRPS

for Walmart roughly ninefold. A Vasicek short rate is fitted to the 3-month Treasury. The few parameters that are assumptions (cash floor, credit facility size where untagged) are labelled as such in the outputs.

3.3 Scenarios, reverse stress and sensitivity

Eight presets (rates +200bp, sales −20%, recession, demand freeze, stagflation, credit crunch, supply-chain shock) are expanded into per-quarter shock paths. Reverse stress testing finds the smallest shock giving a 50% breach probability by bisection under common random numbers. Variance-based Sobol indices [6] over the joint six-dimensional scenario space separate first-order effects from interactions (Fig. 4).

4 Evaluation

4.1 Large-sample probabilistic calibration

The universe is the current S&P 500 outside Financials and Real Estate, plus a handful of former members relevant to 2020. For each of the 375 companies and each yearly cutoff (15 February 2013–2025) we calibrate on point-in-time filings and forecast the next four reported quarters with 2,000 paths, yielding 30,414 forecasts. We score with the CRPS against a seasonal random walk with historical spread and examine probability integral transform (PIT) histograms. Online split-conformal calibration [4, 5] rescales each year's interval half-widths by the empirical quantile of normalized errors from *earlier* cutoffs only.

	Forecasts	CRPS skill	Beats naive	90% cov. raw → conformal	50% cov. raw → conformal	Regime gain
Revenue	15,212	28%	69%	70% → 91%	35% → 51%	0.5%
Cash	15,202	-26%	44%	36% → 91%	20% → 51%	0.1%

Table 1: Out-of-sample evaluation. CRPS skill $= 1 - \mathrm{CRPS}_{\mathrm{twin}}/\mathrm{CRPS}_{\mathrm{naive}}$ (pooled). "Regime gain" is the CRPS reduction from adding the recession regime, holding shocks fixed.

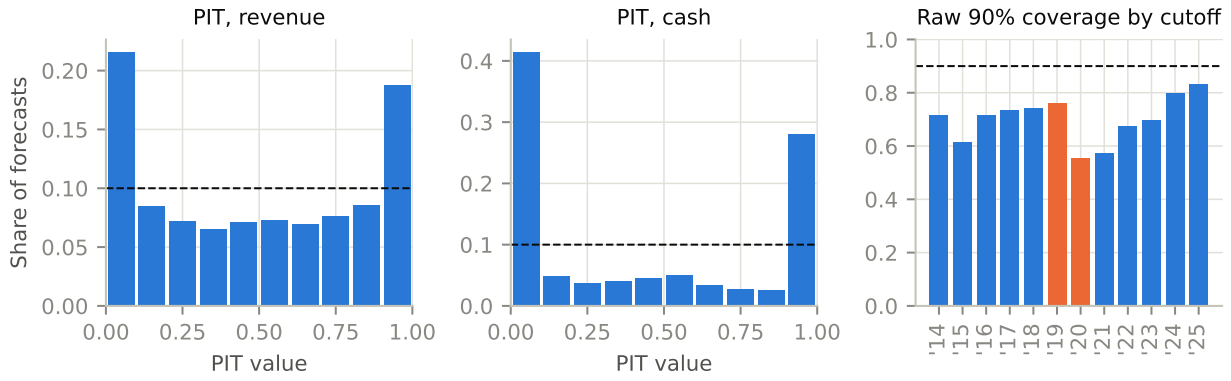


Figure 1: PIT histograms (dashed: uniform) and raw 90% interval coverage by cutoff year. The U-shape shows raw bands are too narrow; the cutoffs whose window contains the 2020 shock (orange) are the worst.

4.2 Did the twin know who would bleed?

We froze every twin at 20 February 2020 and applied the same generic demand-freeze scenario (−70% revenue for three quarters, markets shut for two). The realized outcome is the worst cumulative free cash flow between 2020Q1 and 2021Q2 as a share of pre-shock annual revenue. The twin's stress health score correlates with the realized outcome at $\rho = 0.39$ ($p < 0.001$, $n = 331$, Fig. 2); 16% of the companies in the twin's two most fragile quintiles suffered a severe cash burn (worse than 10% of annual revenue), against 3%

in the two most robust quintiles. Because every company receives the same shock, this measures structural fragility, not exposure: Walmart is fragile to a demand collapse it never suffered.

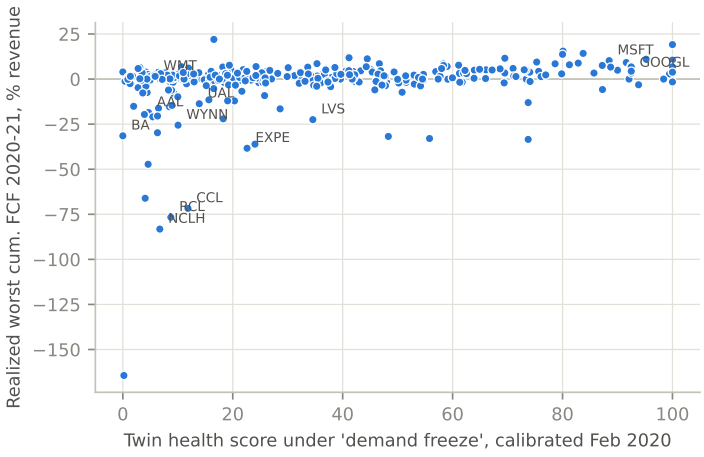


Figure 2: Pre-COVID twin stress health score versus realized worst cumulative free cash flow in 2020–21.

4.3 Structural test

Feeding the realized revenue path into the twin isolates its mechanics from its inability to foresee shocks. For Apple (2020–21) simulated cumulative free cash flow is \$179.1B against \$182.1B actual (error -2%). For Carnival the twin reproduces the direction and timing of the collapse (−\$12.8B vs. −\$17.6B, quarterly correlation 0.66) but underestimates burn, because fixed ship-operating costs and deposit refunds exceeded the calibrated structure (Fig. 3). Boeing and Netflix fail the test for identifiable reasons — program-specific working capital and a content-spending pause — that generic XBRL tags cannot reveal.

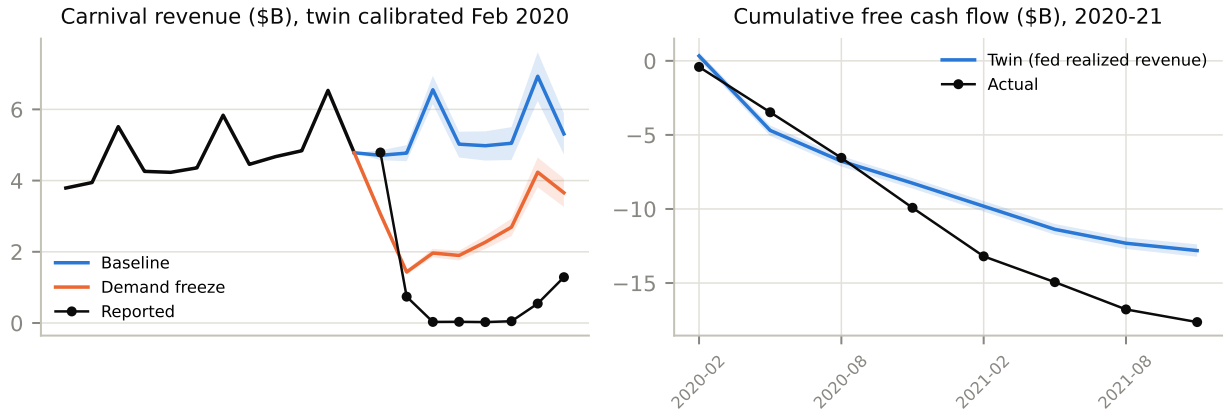


Figure 3: Carnival. Left: revenue forecasts from the twin frozen in February 2020 under the baseline and the demand-freeze scenario, against reported revenue. Right: structural test with realized revenue fed in.

5 Implementation

The Python reference engine (NumPy) performs calibration, backtests and exports; a line-for-line TypeScript port runs in a Web Worker, so the public dashboard needs no server and recomputes 5,000 paths per slider move. Golden runs exported from Python with identical pre-drawn shocks require agreement to 10^{-6} on

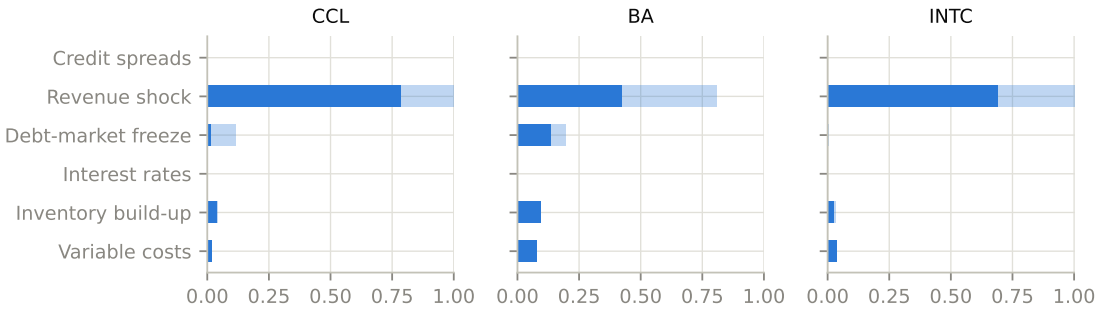


Figure 4: Sobol indices of the health score (solid: first-order; light: total). Large gaps between the two indicate interactions — e.g. a debt-market freeze matters mainly in combination with a revenue shock.

eleven output series. A rule-based analyst turns each company's results into a credit-memo-style note (verdict, evidenced risks, buffers, scenario commentary, watch list) whose figures are taken directly from the simulation output.

6 Limitations

The twin covers non-financial corporates; filers relying on custom taxonomies cannot be parsed with generic tags. Unconditional fans cannot anticipate regime breaks: Carnival's revenue fell about 99%, beyond the -70% preset. Balance-sheet forecasts are weaker than revenue forecasts because financing decisions are discretionary and modelled as calibrated noise. Conformal rescaling corrects average coverage but not conditional coverage in crisis years. Using today's index members introduces survivorship bias: companies that failed before 2026 are absent from the sample, which likely flatters the cross-sectional result.

7 Conclusion

A transparent stock-and-flow twin calibrated from public filings, evaluated as a probabilistic forecaster without look-ahead, is useful less as a predictor of the next quarter than as an instrument for ranking structural fragility and exploring conditional scenarios. All code and data snapshots are open.

AI use statement. Software development, analysis and drafting were carried out with assistance from an AI coding assistant (Claude); the author reviewed the design, results and text.

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